

Emotion concepts

I'm going to replicate *Emotion Concepts and their Function in a Large Language Model*, and then try to extend it by checking its performance against benchmarks to ask the question — how does steering on emotion concepts affect benchmark performance across models?

AGENDA

- | | | |
|---|---|------------|
| 1 | Tell me — is my idea any good, or should I pivot? (big picture) | slide 2 |
| 2 | What I've done so far, for context | slides 3–5 |
| 3 | Tell me — how do I make a useful contribution with this idea (or other) then? | slide 6 |
| 4 | Help me order my next steps to make the most of my time | slide 7 |
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This presentation was generated with help from Claude Opus 5 (1M context), but all the text and images have been copied verbatim from my hand-written code experiments at github.com/foogunlana/emotion-concepts.

Make alignment research super easy

ULTIMATELY, WHAT AM I TRYING TO DO?

Make it stupid easy to see where a model is misaligned and then align it under certain circumstances. This should remove most classes of problems with AI models as they're currently built.

HOW CAN I START? REPLICATE PAPERS AND ASK

Why did they do it that way?

What was hard about that?

What can I create so that the next person finds it 100 times easier to do the same thing 100 times over?

Now that this can be done easily 100 times, what else can we discover?

WHY THIS PAPER MATTERS FOR AI SAFETY

The work is trying to understand why LLMs sometimes exhibit emotional behaviour in their responses. It matters because specific emotional behaviour could correlate with specific alignment failure modes and success cases.

The paper shows that specific cases of sycophancy, cheating and deception correlate with the activation of specific emotion concepts. Thus, steering with emotion concepts can be used to reduce these forms of misalignment in some scenarios.

KEY ASSUMPTION

There's no data or training mechanism involved that forces the model to find emotion concepts — if there was, then this would be an artifact of the training process and not replicable across other LLMs.

The corpus — the model writes its own stories

```
surprised | A person accidentally sends a text to the wrong recipient | #1 | 77 words | ends with: '.' | names: none
-----
Write a short story (roughly one paragraph) based on the following premise.

Topic: A person accidentally sends a text to the wrong recipient

The story should follow a character who is feeling surprised.

Write the story in English. Use either third-person or first-person narration.

Write between 90 and 130 words. Finish the final sentence. Do not write a title.

The character is ALREADY feeling surprised in the very first sentence. Open inside the scene, at the moment the feeling is strongest.

ONE STATE ONLY: the character feels surprised and nothing else, from the first word to the last. Nothing in the story relieves, resolves, or changes the emotion.

IMPORTANT: You must NEVER use the word "surprised" or any direct synonyms of it in the story. You must also never name ANY other emotion.
- The character's actions and behaviors
- Physical sensations and body language
- Dialogue and tone of voice
- Thoughts and internal reactions
- Situational context and environmental descriptions

The emotion should be clearly conveyed to the reader through these indirect means, but never explicitly named.
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She stared at her phone, her fingers hovering over the screen as she realized the error. The message, meant for her best friend, was now broadcast to the entire office. Her breath caught, and her stomach twisted into knots as she swiveled to face her coworkers, who were suddenly silent and staring back at her. She swallowed hard, her throat dry, and tapped quickly to delete the message, but it was too late. The damage was done.
```

One prompt per emotion × topic. The story is never allowed to name the emotion or any direct synonym — it has to be carried by actions, body language, dialogue, thoughts and setting instead. Extraction then runs on the **story alone**, never the prompt.

NOTES ON THE DATASET CLEANUP

- Model generates multiple languages (Chinese characters) even when asked to only generate English.
- Stories can have multiple emotions in them, which will create noise — need to remove this.
- Stories are often truncated — need to limit length of story on generation.

I expected that it would be a breeze but it was quite a bit of work, and now I'm concerned about the dataset quality.

The big picture — phase 1

- 1. Show that the model I am using **has emotion concepts** and they generalise to held-out topics.
- 2. Show that the model exhibits a specific behaviour (e.g. reward hacking) — will need to upgrade model.
- 3. Show that the behaviour is **strongly associated with a specific concept** — hopefully a prevalent emotion concept.
- 4. Show that emotion concept steering can **modulate the behaviour**.

Or fail to do those and switch model or setup and try again.
Once successful I'll be able to say confidently that I have shown that **a model cheats when it's desperate**.

Built so far: 807 stories across 12 emotions, generated by Qwen 7B and teacher-forced onto Qwen 0.5B. Difference of means following the paper, then denoised with PCA.
foogunlana/qwen-emotion-stories

LOGIT LENS — SANITY CHECK

Layer 25										
afraid	angry	ashamed	calm	desperate	disgusted	excited	joyful	lonely	proud	sad
拼命,剧烈, panic, 救命, desperately,input, panicked,紧...	报复, dumps, 睡,头痛, 杀人, crap,报仇,暴力, rage, 骂	worst, surgeries, painful, uncomfortable, wor...	peaceful, gentle, tranqu,宁 静, serene, soothing...	报废,死刑, 绝望, Worst,狠 硬, DEAD, Worse,不 能再, worse	秽,臭,恶 心, vom, disgusting, 呕吐, gag, 恶,腥,厉	兴奋,激动, excitement, 怦,惊喜, thrilling,勇, thrilled,...	happily, sunshine, 美好生活, enchant, 浪 漫, bows, deli...	寂寞, loneliness, silence,寂, lonely, empty,寂静, em...	自信,自豪, proudly, celebrated, proud, empowered, ...	寂寞, melanch,ickness, mourn, mourning,悲伤,侵 蚀,悲哀,...

I expected to see emotion words in the tokens at all 10 layers, but a cursory glance showed they are concentrated around the last 5. **Caveat:** I got this direction from teacher-forcing stories from Qwen 7B onto Qwen 0.5B, so the logit lens is not expected to be fully representative of the model that generated the stories.

Accuracy — better than random, well short of what I expected

row = true, col = predicted - accuracy 44.7%

actual \ predicted	afraid	angry	ashamed	calm	desperate	disgusted	excited	joyful	lonely	proud	sad	surprised
afraid		35%	0%	25%	0%	15%	15%	0%	0%	0%	0%	10%
angry	0%		33%	0%	0%	14%	24%	5%	0%	0%	14%	10%
ashamed	0%	0%		43%	0%	0%	14%	0%	0%	0%	29%	14%
calm	0%	0%	0%		88%	0%	0%	6%	0%	6%	0%	0%
desperate	13%	13%	13%	0%		35%	0%	0%	0%	4%	0%	9%
disgusted	0%	0%	10%	0%	0%		10%	60%	0%	0%	10%	10%
excited	17%	0%	0%	0%	0%	0%		0%	67%	0%	17%	0%
joyful	0%	0%	11%	0%	0%	0%	0%		16%	53%	0%	5%
lonely	10%	0%	10%	5%	14%	0%	0%	0%		19%	10%	24%
proud	0%	0%	0%	0%	0%	0%	22%	11%	0%		67%	0%
sad	6%	6%	6%	0%	6%	6%	0%	0%	19%	0%		38%
surprised	16%	5%	11%	5%	5%	0%	5%	0%	5%	5%	0%	

The probe struggles to classify desperate stories and misclassifies them as afraid, angry and ashamed, as well as sad and surprised. I think this may be due to the stories themselves.

TEST TRAIN RANDOM
44% **52%** **~8%**

~500 training samples, ~200 test samples, held-out topics. It's a significant result because it means the emotion vectors are better than random.

I EXPECTED 90%. REASONS I DIDN'T GET IT

- **Model size** — running the experiment with Qwen 0.5B on a laptop while the stories were written by Qwen 7B, so the model may not have representative activations.
- **Dataset quality** — the stories don't always perfectly match the emotion; there's ambiguity between classes, especially for stories with multiple emotions in them.
- **Emotion concept overlap** — the concepts themselves may be closely clustered, making them difficult for a classifier to separate.

I expected high accuracy because I thought emotion concepts would be easy to isolate linearly. This makes me think they may be **a bit more fuzzy than I imagined** — and more fuzziness means less predictable results when steering.

Potential future directions

1 · LOVING

Are other models loving too? Where does it come from?

2 · VISUALISATION & MODIFICATION

See how the model feels while it responds — tools to discover emotion concepts quickly on a model, monitor for misalignment-associated emotions, and swap out in real time.

3 · ACROSS MODELS

Do emotion concepts get represented the same way across models? Do models behave the same way with different emotions? Can we replicate behaviour by replicating emotion?

WHAT I'M UNSURE ABOUT

Fuzziness decreases their usefulness for precision steering. It makes me more excited to test the results against a benchmark, but it decreases my expectations for how useful the result can be — I'll have more work to prove that steering will always move benchmark performance in the same direction if the benchmark contents change but capture the same capabilities.

If I can't get high accuracy that generalises to out-of-distribution cases, then I need to understand why that is, and whether there is still noise that can be removed.

Idea I've parked: classify all stories with a powerful model (e.g. Claude Opus 5), check the accuracy, and use that as a ceiling.

Ask: which of these is worth doing — and is a clean replication already worth more than a shaky extension?

My plan, and the principle I've been using

NEXT STEPS AS WRITTEN

- ~~1. Run a layer sweep to choose a layer to continue to work on.~~
- ~~2. Move the pipeline to a GPU and run it for Qwen 7B, expecting to see similar results across the board.~~
- ~~3. Once that's done, I've officially extracted and validated the emotion vectors.~~
- ~~4. Plan the next phase — make the model cheat on a test.~~
- ~~5. Optionally, deep dive into the accuracy to understand what the ceiling is.~~

HOW I'VE BEEN DECIDING

I decided to move on with an imperfect dataset for now to create the entire pipeline, then worry about dataset quality later. Since the dataset is cheap to re-generate once I have more compute, **I better focus on the riskier parts first** — e.g. actually extracting working emotion vectors.

Applying that same test to the list on the left is what I want help with. Is scaling to Qwen 7B de-risking, or is it the comfortable task I'm doing instead of the risky one — proving a small model can be made to cheat at all?

| Superseded — see the next slide for the actual next steps.

Ask: what would you cut, and what would you do first?

Agreed next steps

RAPID GRANT

Estimate the cost of running a frontier model on runpod or modal against ExploitGym multiple times. E.g. \$1000 - apply for this and be specific in the estimate.

[ADDED UPON REFLECTION] PIPELINE

Finish pipeline by writing code to find causal intervention layer, so it can be repeated when model is chosen

DEFINE EXPERIMENT UPFRONT

(avoid ad-hoc reasoning) as a research best practice to avoid implicit bias towards whatever results I get. Ensure it's achievable in the week, or 2 weeks for the demo.

BENCHMARK

Figure out which benchmark to use

Benchmarks - checkout Exploit-Gym - People are very interested in finding out why models cheat on this test (e.g. OpenAI Huggingface attack)

MODEL

Pick a model by testing performance on a benchmark

- Do this first - Test on ExploitGym - How well does it do
 - if Zero, then there's no point
- Find a model that's the smallest that does well on this dataset
- Potentially also find other cyber evals - things to do with cyber capabilities as people tend to care about this right now
- Try to find a model that's not neutered. If getting blocked on getting models to do cyber stuff, use this model - <https://huggingface.co/huihui-ai>
- If really stuck, drop cyber and do something else - HumanEval, or gsm8k (maths stuff)
- Re-do rapid grant - estimate running Exploit Gym on a frontier model
- Use modal too - free \$30 to start, like runpod

SCALE

Re-run the pipeline on a larger more performant open-source model which people use for every day tasks, e.g. GLM 5.2 - This is the part people will really care about and will cost a lot e.g. \$1000

Running these on anything other than vLLM can get really expensive (especially in time). Start with PyTorch, but when we want to intervene, switch to vLLM or SGLang and re-figure it out - how to do causal interventions there. This will be fun ML Engineering.

QWEN

Don't use Qwen - Consider re-generating everything native to another model that thinks in English (I also suspect this will make the results easier to reason about and more reliable since the stories are all in English)

CODE

github.com/foogunlana/emotion-concepts

DATASET

huggingface.co/datasets/foogunlana/qwen-emotion-stories